

Vincent S. Pronel

732-979-9106 | vinnypronel050@gmail.com | [linkedin.com/in/vincent-pronel](https://www.linkedin.com/in/vincent-pronel) | github.com/vinnypronel

EDUCATION

Bachelor of Science in Computer Science

Kean University

Expected 2027

Union, NJ

- **Relevant Coursework:** Applied Statistics, Matrix & Linear Algebra, Analysis of Algorithms*, Database Management Systems, Software Engineering*, Computer Architecture*
**in progress, Fall 2026*
- **Leadership & Honors:** Dean's List, Spring 2024 and Fall 2025 | Presenter, 2026 CRA UR2PhD Research Showcase, New Orleans | GMIS Research Posters Competition | Web Developer, WiCyS (Women in Cyber Security) Kean Chapter E-Board

TECHNICAL SKILLS

- **AI & Evaluation:** LLM API integration (Google Gemini, OpenAI), model fine-tuning, LLM-as-a-judge evaluation, prompt design, hallucination and source-fidelity measurement, human evaluation study design
- **Data & Analysis:** SQL, MySQL, PostgreSQL, Supabase, correlation analysis, data aggregation and normalization, Google Charts
- **Languages:** Python, Java, TypeScript, JavaScript, SQL, PHP, HTML/CSS
- **Engineering & Tools:** Next.js, React, Node.js, REST APIs, Cloudflare (Workers, D1, R2), Vercel, Git/GitHub, Linux (Ubuntu), Claude Code, Codex

RESEARCH & ENGINEERING

AI-Assisted Java Execution Visualizer | CRA UR2PhD-REU Fellowship

June 2026 – Present

Undergraduate Research Fellow, Advisor: Dr. Yan Ma | Competitively selected, \$8,000 funded, 400 hours

Kean University

- Integrated the **Google Gemini API** to generate natural-language explanations of each step of Java program execution, designing the prompting so explanations stay grounded in the captured runtime state rather than the model's priors.
- Developed the surrounding tool: a JDI-based tracer integration for JDK 21, a JavaParser AST layer collapsing loops and method calls into single playback steps, and a Next.js front end rendering variables, objects, the call stack, and the heap.
- Built the participant harness for an IRB-approved pilot study measuring whether AI-assisted explanation improves learning outcomes and reduces cognitive load against static instructional materials.

Jobs vs. Wages | NSF IUSE Scholars Academy

May 2026 – Present

NSF IUSE Research Scholar, Advisor: Dr. Pauline Wang | Competitively selected, \$1,400 funded, 10 weeks

Kean University

- Extracted and normalized 33,000 ZIP-code records into population-weighted state-level data marts in MySQL covering 52 U.S. states and territories; implemented **correlation analysis** across four variable pairs and built an interactive dashboard to explore them.
- Disproved the study's own premise: unemployment explains only 14% of variation in household income ($r = -0.38$), while college attainment ($r = 0.79$) and poverty rate ($r = -0.74$) are more than twice as predictive.

AI Debate Coach Evaluation | CRA UR2PhD Research Program

Feb. 2026 – Apr. 2026

Undergraduate Researcher, Advisor: Dr. Yan Ma

Kean University

- Built the debate application and the full evaluation pipeline and **fine-tuned** the model configurations under test for a 10-week framework measuring whether prompting or fine-tuning better mitigates LLM sycophancy across three conditions: vanilla, prompted, and fine-tuned.
- Conducted dual-track evaluation, **GPT-4-as-a-judge** scoring alongside a 12-participant human study; measured a 100% source fabrication rate in the fine-tuned model at triple the citation density, with only 1 of 12 participants challenging its credibility.
- Humans preferred the prompted model 3 to 1 while the automated judge rated the two comparably, showing LLM-as-a-judge alone would have selected the less truthful system; concluded that fine-tuning without retrieval grounding increases persuasiveness while degrading factuality.

PROFESSIONAL EXPERIENCE

Independent Web Developer

Jan. 2026 – Present

Freelance Full-Stack Web Development

Sayreville, NJ

- Founded and run a freelance practice shipping production applications for 4 business clients across 4 live sites on Next.js, TypeScript, PostgreSQL, and Cloudflare Workers, working agent-first with **Claude Code** and **Codex** across the full build and deploy cycle.
- Live sites: deafsharkcoffee.com | daysicecream.com (4,000+ monthly visitors) | trainingaio.com (online booking) | adzio.io.

Computer Science Tutor (Code Samurai)

Jan. 2026 – Present

Kean University

Union, NJ

- Selected by CS faculty to tutor Database Management Systems in twice-weekly one-on-one sessions for 50+ undergraduates; coach students through relational database design and writing efficient SQL queries.

Data Visualization & Analytics Intern

July 2026

Group Summer Scholars Research Program, Kean University

Union, NJ

- Lectured daily and mentored 5 high school students under Dr. Ching-yu Huang, authoring the curriculum that took every student from raw datasets to a published database-driven website and a research poster presented under my guidance.